

TIKHAVIN, Russian Federation

WMO: 260940

Lat: 59.650N

Lon: 33.550E

Elev: 61

StdP: 100.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.8	-24.0	-30.4	0.2	-27.7	-26.4	0.3	-23.9	6.3	-1.8	5.8	-1.7	0.6	60	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.5	19.9	26.3	18.7	24.4	17.7	20.9	26.6	19.9	24.6	18.8	23.2	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	13.9	23.3	18.1	13.1	22.0	17.1	12.3	21.1	60.7	26.8	57.1	24.6	53.7	23.4	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.2	4.6	-31.3	31.2	4.5	2.1	-34.5	32.7	-37.1	33.9	-39.6	35.1	-42.8	36.6		
			-31.4	22.3	4.4	1.4	-34.6	23.3	-37.2	24.1	-39.7	24.8	-42.9	25.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-7.9	-8.0	-2.9	4.3	10.4	14.9	18.0	15.7	10.6	4.5	-1.4	-5.5	(6)
(7)		DBStd	10.46	7.47	7.54	5.04	4.58	4.69	3.84	3.59	3.42	3.84	4.13	5.49	7.07	(7)
(8)		HDD10.0	2733	553	504	401	181	53	4	1	2	38	174	343	480	(8)
(9)		HDD18.3	5133	812	738	659	421	249	117	49	97	232	427	593	738	(9)
(10)		CDD10.0	711	0	0	0	9	65	151	248	177	57	4	0	0	(10)
(11)		CDD18.3	68	0	0	0	0	3	13	38	13	1	0	0	0	(11)
(12)		CDH23.3	674	0	0	0	3	39	154	350	122	6	0	0	0	(12)
(13)		CDH26.7	164	0	0	0	0	6	36	97	25	0	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.2	2.2	2.3	2.2	2.3	2.1	1.9	1.9	2.0	2.2	2.3	2.4	(14)
(15)	Precipitation	PrecAvg	762	61	42	42	41	59	82	81	87	62	67	71	67	(15)
(16)		PrecMax	999	93	76	76	68	108	172	241	175	103	110	107	123	(16)
(17)		PrecMin	500	15	8	7	17	17	26	27	10	17	30	4	31	(17)
(18)		PrecStd	108	20	18	18	15	24	34	46	42	23	23	23	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	4.4	11.7	22.7	27.1	30.2	32.4	30.0	24.2	16.5	9.2	7.3	(19)
(20)		MCWB	2.8	3.0	6.0	13.4	16.8	19.5	21.6	20.8	18.1	13.0	7.9	6.3	(20)	
(21)		2%	DB	2.2	2.9	7.4	18.9	24.0	27.2	29.2	26.2	21.0	13.0	7.7	5.0	(21)
(22)		MCWB	1.5	2.1	3.6	11.2	15.2	19.0	20.6	18.9	16.2	11.2	6.9	4.4	(22)	
(23)		5%	DB	1.4	2.0	4.9	15.5	21.5	24.8	27.0	24.2	18.4	11.5	6.3	3.6	(23)
(24)		MCWB	0.9	1.2	2.2	9.3	13.8	17.3	19.4	17.9	14.7	10.2	5.7	3.0	(24)	
(25)		10%	DB	0.6	1.0	3.3	12.5	19.2	22.6	24.9	22.3	16.5	10.3	5.0	2.1	(25)
(26)		MCWB	0.2	0.4	1.6	7.4	12.4	16.0	18.5	17.0	13.6	9.3	4.4	1.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.6	6.7	14.1	19.5	21.0	22.7	22.4	19.0	13.3	8.3	6.5	(27)
(28)		MDCB	3.4	4.0	11.3	21.7	25.0	27.4	29.7	28.1	23.0	15.6	8.9	7.2	(28)	
(29)		2%	WB	1.8	2.2	4.2	11.8	16.5	19.8	21.4	20.2	16.9	11.6	7.0	4.5	(29)
(30)		MDCB	2.1	2.6	6.2	17.7	21.2	26.1	27.7	24.4	19.8	12.6	7.4	4.9	(30)	
(31)		5%	WB	1.0	1.3	2.8	10.1	14.7	18.4	20.3	18.8	15.3	10.5	5.7	3.1	(31)
(32)		MDCB	1.3	1.8	4.2	14.4	19.6	23.4	25.2	22.8	17.6	11.4	6.3	3.5	(32)	
(33)		10%	WB	0.4	0.5	1.9	8.1	13.3	17.0	19.4	17.6	14.1	9.4	4.5	1.7	(33)
(34)		MDCB	0.7	1.0	3.1	11.8	18.2	20.6	23.7	21.1	16.3	10.3	5.0	2.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.3	6.9	8.3	10.2	11.4	10.3	9.9	9.8	8.2	5.2	4.1	4.6	(35)	
(36)		5% DB	MCDBR	3.3	3.9	9.8	16.0	15.4	14.0	13.2	13.3	11.2	7.2	4.4	3.5	(36)
(37)		MCWBR	3.4	3.5	6.8	9.0	7.8	6.5	5.6	6.6	6.5	5.2	4.1	3.4	(37)	
(38)		5% WB	MCDBR	3.3	3.8	7.3	14.3	13.1	11.9	11.7	11.1	9.6	6.4	4.2	3.5	(38)
(39)		MCWBR	3.4	3.6	5.3	8.3	7.5	6.0	5.4	6.1	5.9	4.9	4.1	3.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.311	0.331	0.373	0.367	0.362	0.392	0.387	0.355	0.332	0.300	0.253	(40)	
(41)		taud	2.084	2.087	2.140	2.253	2.398	2.436	2.366	2.369	2.440	2.391	2.200	2.056	(41)	
(42)		Ebn at Noon	495	666	787	823	861	871	830	805	773	664	462	368	(42)	
(43)		Edh at Noon	47	81	105	113	105	103	108	100	79	61	43	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.30	0.93	2.35	3.73	5.04	5.37	5.12	4.03	2.42	0.97	0.29	0.16	(44)	
(45)		RadStd	0.05	0.22	0.31	0.46	0.47	0.38	0.65	0.40	0.28	0.15	0.04	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+0.64	N/A	N/A	N/A	N/A	N/A	-189	-228	N/A	N/A	(47)

Nomenclature: See separate page